



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No : 7

TITLE :Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.



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Code and Output Screenshots

```
J main3.java > Language Support for Java(TM) by Red Hat > main3
1  class Animal {
2      void eat() {
3          System.out.println("This animal eats food.");
4      }
5  }
6
7  class Cat extends Animal {
8      void meow() {
9          System.out.println("Cat meows!");
10     }
11 }
12
13 public class main3 {
14     Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16         Cat d = new Cat();
17         d.eat();
18         d.meow();
19     }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text)

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac main3.java && java main3
This animal eats food.
Cat meows!

[Done] exited with code=0 in 0.475 seconds
```



```
1 class S1 {
2     String name;
3
4     S1(String name) {
5         this.name = name;
6     }
7
8     void display() {
9         System.out.println("Name: " + this.name);
10    }
11 }
12
13 public class Main2 {
14     Run main | Debug main | Run | Debug
15     public static void main (String[] args) {
16         S1 s = new S1("Pushyamitra Shunga");
17         s.display();
18     }
19 }
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text)

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main2.java && java Main2
Name: Pushyamitra Shunga

[Done] exited with code=0 in 0.485 seconds

Conclusion

This program demonstrates the use of inheritance and interfaces in Java using a real-world example. Inheritance helps reuse existing code, while interfaces define common behaviors that different classes can implement. These concepts make programs more organized, reusable, and easier to maintain.



Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
Ex1.java > Java > Ex1
1  class Shape {
2      void display() {
3          System.out.println("New shape created!");
4      }
5  }
6
7  class Circle extends Shape {
8      double area (double r) {
9          return 3.14 * r * r;
10     }
11 }
12 class Rectangle extends Shape {
13     double area (double l, double b) {
14         return l * b;
15     }
16 }
17 public class Ex1 {
18     Run main | Debug main | Run | Debug
19     public static void main (String[] args) {
20         Circle c = new Circle();
21         Rectangle r = new Rectangle();
22
23         c.display();
24         System.out.println("Area of Circle : " + c.area(9.6));
25         r.display();
26         System.out.println("Area of Rectangle : " + r.area(6.7, 4.56));
27     }
28 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...)

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Ex1.java && java Ex1
New shape created!
Area of Circle : 289.38239999999996
New shape created!
Area of Rectangle : 30.552

[Done] exited with code=0 in 0.513 seconds
```



2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
1 interface Product {
2     void showPrice();
3 }
4 class Item {
5     String name;
6     Item(String name) {
7         this.name = name;
8     }
9     void display() {
10        System.out.println("Product: " + name);
11    }
12 class Electronic extends Item implements Product {
13     Electronic(String name) {
14         super(name);
15     }
16     public void showPrice() {
17         System.out.println("Price for electronic item: Rs. 25000");
18     }
19 class Clothing extends Item implements Product {
20     Clothing(String name) {
21         super(name);
22     }
23     public void showPrice() {
24         System.out.println("Price for clothing item: Rs. 1500");
25     }
26 class Grocery extends Item implements Product {
27
28     Grocery(String name) {
29         super(name);
30     }
31     public void showPrice() {
32         System.out.println("Price for grocery item: Rs. 120");
33     }
34 public class Ex2 {
35     Run main | Debug main | Run | Debug
36     public static void main(String[] args) {
37         Electronic e = new Electronic("Laptop");
38         Clothing c = new Clothing("T-Shirt");
39         Grocery g = new Grocery("Rice");
40         e.display();
41         e.showPrice();
42         c.display();
43         c.showPrice();
44         g.display();
45         g.showPrice();
46     }
47 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Ex2.java && java Ex2
Product: Laptop
Price for electronic item: Rs. 25000
Product: T-Shirt
Price for clothing item: Rs. 1500
Product: Rice
Price for grocery item: Rs. 120
```



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